Recommendations — pancreatic-recurrent-kras-g12r-m8f3

Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references

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Recommendations — pancreatic-recurrent-kras-g12r-m8f3

Profile snapshot

- **Primary site:** pancreas
- **Histology:** pancreatic adenocarcinoma
- **Stage:** recurrent
- **ECOG:** 1
- **Age band:** 50-59
- **Sex:** unknown
- **Biomarkers:** KRAS G12R mutated; TP53 inactivating mutation; CDKN2A loss; CCND3 alteration; MSI status MSS; TMB 4.1 mut/Mb
- **Efficacy/toxicity weight:** 0.8

10 ranked therapeutic options across 3 targetable features. Biomarker workup steps live in the standalone Target validation paths report.

KRAS G12R-targeting interventions

1. daraxonrasib (RMC-6236) 300 mg PO daily monotherapy on NCT05379985 (RMC-6236-001)

Likelihood of effect: High in 2L RAS G12 PDAC at 300 mg — ORR 35%, mPFS 8.5 mo, mOS 13.1 mo (Wolpin NEJM 2026, n=26). G12R is inside the pooled estimand and preclinical activity is allele-replicated.

Toxicity burden: Moderate (rash 91% any-grade / G3+ 8%; diarrhea 48% / G3+ 2%; stomatitis 31% / G3+ 3%; G3+ TRAE 30% overall)

Counter-productive MoA:Low (Residual wild-type-RAS pharmacology window is tight in some tissues; KPC-model relapse linked to MYC amplification (Wasko 2024))

Overall:Only published meaningful effect size on KRAS G12R PDAC; recruiting trial with fully mapped AE distribution and matched mechanism. Phase 3 confirmation pending.

Key references: PMID 42090791 · NCT05379985 · PMID 38589574 · PMID 38593348

2. PF-07934040 pan-KRAS small-molecule inhibitor on NCT06447662 (Part 2a Cohort A1, 2L+ PDAC monotherapy)

Likelihood of effect: Moderate — class extrapolation from daraxonrasib (PMID 42090791); no PF-07934040 efficacy readout yet. G12R explicit in eligibility (NCT06447662).

Toxicity burden: Low (no published AE data yet; class-effect rash + GI expected from pan-KRAS chemistry; G2+ neuropathy is an enrollment exclusion)

Counter-productive MoA:Low (Same pan-KRAS class window as daraxonrasib; mechanism-level risk has no persona dissent on this row.)

Overall:Backup-class pan-KRAS lane with explicit G12R eligibility; useful if the daraxonrasib slot is unavailable, but no published efficacy yet anchors the rank.

Key references: NCT06447662

3. daraxonrasib (RMC-6236) + mFOLFIRINOX or gemcitabine/nab-paclitaxel on NCT06445062 (RMC-GI-102)

Likelihood of effect: Moderate-to-high if RAMP-205 dose-level-1 (5/6 PR, ORR 83%, n=6) survives N — exact 95% CI on 5/6 runs 36-100%. Daraxonrasib + chemo PDAC-cohort efficacy not yet public.

Toxicity burden: High (rash + GI from pan-RAS stacked with chemo cytopenias and neuropathy; no published dedicated combination-safety table)

Counter-productive MoA:Moderate (Cumulative on-target wild-type-RAS exposure plus chemo myelosuppression — conservative dissented on the missing combination-safety publication.)

Overall:Higher-ceiling combination lane that matches the treat-to-remission preference, but no published dedicated PDAC combination-safety data yet — contingent on sponsor cohort openness and an AE-distribution readout.

Key references: NCT06445062 · PMID 42090791 · doi:10.1200/JCO.2024.42.16_suppl.4140

4. zoldonrasib (RMC-7977) + ivonescimab (PD-1xVEGF bispecific) on NCT07397338

Likelihood of effect: Low for clinical effect at present — zero published PDAC clinical data; preclinical KPC ceiling is high (Wasko Nature 2024) but allele-mismatched (G12D not G12R).

Toxicity burden: Moderate (pan-RAS rash + GI stacked with PD-1xVEGF immune-mediated AEs; no published PDAC combination safety table)

Counter-productive MoA:Moderate (Diagnostic ambiguity between class rash and immune-rash complicates AE attribution; conservative dissented on this mechanism-level overlap.)

Overall:Highest-ceiling preclinical option the patient could touch, but zero published clinical PDAC data and three persona dissents on evidence, toxicity, and guideline-fit.

Key references: PMID 38588697 · NCT07397338 · PMID 38589574

5. ELI-002 7P amphiphile mKRAS peptide vaccine on NCT05726864 (AMPLIFY-7P) — patient sits outside the MRD-positive resected enrollment window

Likelihood of effect: Low for this clinical state — designed for MRD-positive resected disease, not overt recurrence; AMPLIFY-201 read in the MRD window (Wainberg Nat Med 2025).

Toxicity burden: Low (zero DLTs; no G3+ vaccine-attributed events across AMPLIFY-201 n=25; injection-site reactions dominate)

Counter-productive MoA:Low (T-cell exhaustion / antigen-loss escape on overt-disease bulk are the theoretical concerns; mechanism designed for MRD setting.)

Overall:Cleanest safety in the dossier with G12R-explicit mechanism fit, but the MRD-positive enrollment window forecloses this patient's clinical state.

Key references: NCT05726864 · PMID 38195752 · PMID 40790272

6. avutometinib (RAF/MEK clamp) + defactinib (FAKi) + gemcitabine/nab-paclitaxel on NCT05669482 (RAMP-205)

Likelihood of effect: Moderate-to-high in cross-tumor KRAS-mutant disease (LGSOC ORR 44% Banerjee 2025); PDAC signal is abstract-only at n=6 with 95% CI 36-100%.

Toxicity burden: Moderate (four-agent stack: RAF/MEK class rash, edema, CK / FAKi GI / chemo cytopenias and neuropathy; no published dedicated PDAC combination AE table)

Counter-productive MoA:Moderate (Critic dissented on evidence-quality; cumulative MAPK + stromal + cytotoxic toxicity overlap is the mechanism-level risk to dose intensity.)

Overall:Striking n=6 PDAC abstract signal and KRAS-mutant LGSOC cross-tumor validation, but trial is active not recruiting and the evidence base is one unpeer-reviewed abstract.

Key references: doi:10.1200/JCO.2024.42.16_suppl.4140 · PMID 40644648 · PMID 27376576 · NCT05669482

Pipeline context — not currently enrollable

Other agents in this target class drawn from the case dossier. The patient cannot currently enroll in these trials, but they are included for context on the broader modality landscape.

Intervention	Modality	Phase	Enrolling indication	Recruitment	Trial
ASP3082 (setidegrasib, KRAS G12D-selective degrader) + mFOLFIRINOX or NALIRIFOX	small_molecule	3	First-line KRAS G12D-mutated metastatic PDAC	recruiting	NCT07409272
INCB161734 (KRAS G12D-selective) + chemotherapy	small_molecule	3	Previously untreated KRAS G12D-mutated metastatic PDAC (DAWN-303)	recruiting	NCT07522073
ASP5834 pan-KRAS inhibitor (intravenous)	small_molecule	1	Locally advanced / metastatic solid tumors with KRAS mutations or amplification — NSCLC, PDAC, CRC	recruiting	NCT07094204
MEK inhibitor-based combinations (MEKi + HCQ, MEKi + EGFRi, etc.)	other	n/a	Advanced KRAS G12R altered PDAC	recruiting	NCT05630989

MRTX1133 (KRAS G12D-selective) _(discontinued)_	small_molecule	1/2	Advanced solid tumors with KRAS G12D mutation	terminated	NCT05737706

Evidence in detail

One row per published clinical manuscript supporting each ranked intervention, drawn from the case's clinical-evidence dossier.

daraxonrasib (RMC-6236) 300 mg PO daily monotherapy on NCT05379985 (RMC-6236-001)

Wolpin et al. *New England Journal of Medicine* 2026

- **Disease context:Previously treated advanced RAS-mutated pancreatic ductal adenocarcinoma (PDAC)** (2L+) — RMC-6236-001 (NCT05379985) phase 1/2 PDAC cohort; 168 PDAC patients dosed at 300 mg or lower; second-line RAS G12 subgroup analyzed at the 300-mg dose (n=26). G12X variants enrolled include G12D, G12V, G12R, and Q61H — the patient's G12R is in scope.
- **n:** 168
- **Toxicities:any treatment-related AE** (grade any): 96% (n=168); **any treatment-related AE** (grade >=3): 30% (n=168); **rash** (grade any): 91% (n=127); **rash** (grade >=3): 8% (n=127); **diarrhea** (grade any): 48% (n=127); **diarrhea** (grade >=3): 2% (n=127); **nausea** (grade any): 43% (n=127); **stomatitis** (grade any): 31% (n=127); **stomatitis** (grade >=3): 3% (n=127); **vomiting** (grade any): 31% (n=127); **fatigue** (grade any): 20% (n=127)
- **Efficacy:ORR** 35%; **DoR** 8.2 months; **PFS** 8.5 months; **OS** 13.1 months; **TRAE_rate** 96%; **TRAE_rate** 30%

daraxonrasib (RMC-6236) + mFOLFIRINOX or gemcitabine/nab-paclitaxel on NCT06445062 (RMC-GI-102)

Wolpin et al. *New England Journal of Medicine* 2026

- **Disease context:Previously treated advanced RAS-mutated pancreatic ductal adenocarcinoma (PDAC)** (2L+) — RMC-6236-001 (NCT05379985) phase 1/2 PDAC cohort; 168 PDAC patients dosed at 300 mg or lower; second-line RAS G12 subgroup analyzed at the 300-mg dose (n=26). G12X variants enrolled include G12D, G12V, G12R, and Q61H — the patient's G12R is in scope.
- **n:** 168
- **Toxicities:any treatment-related AE** (grade any): 96% (n=168); **any treatment-related AE** (grade >=3): 30% (n=168); **rash** (grade any): 91% (n=127); **rash** (grade >=3): 8% (n=127); **diarrhea** (grade any): 48% (n=127); **diarrhea** (grade >=3): 2% (n=127); **nausea** (grade any): 43% (n=127); **stomatitis** (grade any): 31% (n=127); **stomatitis** (grade >=3): 3% (n=127); **vomiting** (grade any): 31% (n=127); **fatigue** (grade any): 20% (n=127)
- **Efficacy:ORR** 35%; **DoR** 8.2 months; **PFS** 8.5 months; **OS** 13.1 months; **TRAE_rate** 96%; **TRAE_rate** 30%

ELI-002 7P amphiphile mKRAS peptide vaccine on NCT05726864 (AMPLIFY-7P) — patient sits outside the MRD-positive resected enrollment window

Pant et al. *Nature Medicine* 2024

- **Disease context:Resected PDAC and CRC with ctDNA / tumor-marker MRD positivity after locoregional therapy (AMPLIFY-201)** (adj) — MRD-positive after curative-intent surgery and standard adjuvant therapy; 25 patients (20 PDAC, 5 CRC); KRAS G12D or G12R required. The patient's G12R is in scope, but the patient is overt-recurrent rather than MRD-only.
- **n:** 25

- **Toxicities:dose-limiting toxicity** (grade >=3): 0% (n=25); **injection-site reaction** (grade any): (n=25)
- **Efficacy:biomarker** 84%; **RFS** 16.33 months; **RFS** not reached months; **RFS** 4.01 months; **AE_rate** 0%

Wainberg et al. *Nature Medicine* 2025

- **Disease context:Resected PDAC and CRC with MRD positivity post locoregional therapy (AMPLIFY-201 final follow-up)** (adj) — Final report at 19.7 months median follow-up; n=25 (20 PDAC, 5 CRC); KRAS G12D / G12R targeted.
- **n:** 25
- **Toxicities:** Maintained tolerability at extended follow-up; 71% of evaluable patients induced both CD4 and CD8 KRAS-specific T-cell responses.
- **Efficacy:RFS** not reached months; **RFS** 3.02 months; **OS** not reached months; **OS** 15.98 months; **biomarker** 71%

avutometinib (RAF/MEK clamp) + defactinib (FAKi) + gemcitabine/nab-paclitaxel on NCT05669482 (RAMP-205)

Banerjee et al. *Journal of Clinical Oncology* 2025

- **Disease context:Recurrent low-grade serous ovarian cancer (LGSOC), KRAS-mutant and wild-type (RAMP 201)** (2L+) — Recurrent LGSOC after ≥1 prior systemic therapy; KRAS-mutant ORR pulled separately. Cross-tumor: not PDAC but the load-bearing efficacy precedent for the RAF/MEK + FAK clamp combo in a KRAS-mutant disease.
- **n:** —
- **Toxicities:** Manageable AE profile; most patients continued therapy. Class effects include rash, edema, CK elevation, and reversible ocular events.
- **Efficacy:ORR** 44%; **ORR** 17%; **DoR** 31.1 months; **DoR** 9.2 months; **PFS** 22 months; **PFS** 12.8 months

CDKN2A-loss / MTAP-targeting interventions

1. anvumetostat (AMG 193, MTA-cooperative PRMT5 inhibitor) + daraxonrasib on NCT06360354 (MTAPESTRY-103) — gated on MTAP co-deletion confirmation

Likelihood of effect: Moderate if MTAP is co-deleted — preclinical foundation strong (>70-fold selectivity, Smith 2023). No PDAC clinical readout yet; combination arm covers two features on one trial.

Toxicity burden: Low (no published clinical AE data for the combination; MTA-cooperative class designed to spare the hematologic toxicity of first-generation PRMT5 inhibitors)

Counter-productive MoA:Low (Synthetic-lethal mechanism is clean; cooperative MTA chemistry limits exposure to MTAP-proficient tissue.)

Overall:Dual-feature trial covering KRAS G12R and CDKN2A / MTAP loss on a single regimen — gated on MTAP co-deletion reflex testing.

Key references: NCT06360354 · PMID 26912361 · PMID 26912360 · PMID 37552839

2. palbociclib + MEK/ERK or IGF1R combination via ADOPT PDO-guided platform

(NCT06813079) or off-label combination on CDKN2A-loss / CCND3 rationale

Likelihood of effect: Low for monotherapy (Z1C ORR 4%, O'Hara 2025); preclinical combination synergy in PDAC organoids (Knudsen 2023) has no PDAC clinical translation yet.

Toxicity burden: Moderate (G3+ neutropenia is the class-defining AE; cumulative cytopenia risk when combined with chemo or MEKi)

Counter-productive MoA:Moderate (Single-agent CDK4/6i triggers RB-bypass compensation via cyclin D, MYC, and mTOR — the documented monotherapy escape route that Z1C confirmed.)

Overall:Mechanism-fit on the CDKN2A / CCND3 axis but the closest published basket precedent (Z1C ORR 4%) is actively negative and PDAC combination data are preclinical-only.

Key references: PMID 36346366 · PMID 25156567 · PMID 24986516 · PMID 39437014 · NCT06813079

Pipeline context — not currently enrollable

Other agents in this target class drawn from the case dossier. The patient cannot currently enroll in these trials, but they are included for context on the broader modality landscape.

Intervention	Modality	Phase	Enrolling indication	Recruitment	Trial
palbociclib (CDK4/6 inhibitor) _(basket/biomarker, off-label)_	small_molecule	2	Advanced solid tumors with actionable genomic alteration — basket	recruiting	NCT03297606
palbociclib _(basket/biomarker, off-label)_	small_molecule	2	Rare solid tumors with actionable genomic alteration — platform	recruiting	NCT04423185
BMS-986504 (navlimetostat / PRMT5 inhibitor) ± daraxonrasib, immunotherapy, or chemotherapy _(basket/biomarker)_	small_molecule	2	Solid tumors (GBM, melanoma, NSCLC, PDAC) with homozygous MTAP deletion	not yet recruiting	NCT07492680

Evidence in detail

One row per published clinical manuscript supporting each ranked intervention, drawn from the case's clinical-evidence dossier.

palbociclib + MEK/ERK or IGF1R combination

O'Hara et al. *Clinical Cancer Research* 2025

- **Disease context:Histology-agnostic advanced solid tumors with CDK4 or CDK6 amplification (NCI-MATCH subprotocol Z1C)** (2L+) — 38 eligible/treated patients with confirmed CDK4 or CDK6 amplification (≥7 copies). 25 patients in the primary analysis cohort. Pancreatic cancer was not the dominant histology; the read-out is for the CDK4/6 amplification call, not CDKN2A loss.

- **n:** 38
- **Toxicities:neutropenia** (grade >=3): (n=38)
- **Efficacy:ORR** 4%; **PFS** 2.0 months; **OS** 8.8 months

Germline BRCA / HRD-targeting interventions

1. olaparib 300 mg PO BID maintenance after platinum induction — POLO regimen (NCT02184195), gated on germline BRCA1/2 (or PALB2) panel + ≥16 weeks platinum disease control

Likelihood of effect: High when both gates open — PFS HR 0.53 (Golan 2019 NEJM POLO). Interim OS HR 0.91 means OS benefit not yet demonstrated.

Toxicity burden: Moderate (G3+ AEs 40% olaparib vs 23% placebo; AE discontinuation 5%; MDS / AML class risk ~1.5% on prolonged exposure)

Counter-productive MoA:Low (Synthetic-lethal mechanism well-characterized; reversion mutations restoring BRCA function are the documented resistance route.)

Overall:Only RCT-grade option in the dossier — high-confidence when germline BRCA/PALB2 confirms and platinum induction succeeds, but two sequential gates remain locked.

Key references: PMID 31157963 · NCT02184195 · PMID 33970687

2. rucaparib 600 mg PO BID maintenance — gated on germline or somatic BRCA1/2 or PALB2 + platinum induction (Reiss 2021 regimen)

Likelihood of effect: Moderate when gates open — Reiss 2021 single-arm n=46 mPFS 13.1 / mOS 23.5 in platinum-sensitive BRCA / PALB2 PDAC; not RCT-grade.

Toxicity burden: Moderate (PARP class effects mirroring olaparib; MDS / AML class risk on prolonged exposure)

Counter-productive MoA:Low (Same synthetic-lethal mechanism as olaparib; reversion mutations restoring BRCA function are the documented resistance route.)

Overall:PARP-class backup that widens the biomarker net beyond germline BRCA to include somatic BRCA / PALB2; same two locked gates as olaparib, phase 2 single-arm evidence.

Key references: PMID 33970687

Evidence in detail

One row per published clinical manuscript supporting each ranked intervention, drawn from the case's clinical-evidence dossier.

olaparib 300 mg PO BID maintenance after platinum induction — POLO regimen (NCT02184195), gated on germline BRCA1/2 (or PALB2) panel + ≥16 weeks platinum disease control

Golan et al. *New England Journal of Medicine* 2019

- **Disease context:Germline BRCA1/2-mutated metastatic PDAC, maintenance after first-line platinum-based chemotherapy (POLO)** (maintenance) — 154 randomized; gBRCA1/2 pathogenic variant; no progression after ≥16 weeks of first-line platinum. The patient has no germline data yet and was treated with adjuvant FOLFIRI (no platinum exposure), so both biomarker and platinum-exposure gates are open.
- **n:** 154
- **Toxicities:any treatment-related AE** (grade >=3): 40% (n=92); **any treatment-related AE** (grade >=3): 23% (n=62); **discontinuation for AE** (grade any): 5% (n=92)
- **Efficacy:HR_PFS** 0.53 HR (95% CI 0.35-0.82); **PFS** 7.4 months; **PFS** 3.8 months; **HR_OS** 0.91; **OS** 18.9 months; **OS** 18.1 months; **ORR** 23.1%; **ORR** 11.5%

Reiss et al. *Journal of Clinical Oncology* 2021

- **Disease context:Platinum-sensitive advanced PDAC with germline or somatic BRCA1/2 or PALB2 pathogenic variant** (maintenance) — 46 enrolled, 42 evaluable; germline BRCA1 n=7, BRCA2 n=27, PALB2 n=6; somatic BRCA2 n=2. Broader than POLO by including PALB2 and somatic variants.
- **n:** 46
- **Toxicities:anemia** (grade any): 74% (n=46); **anemia** (grade >=3): 22% (n=46); **nausea** (grade any): 48% (n=46); **nausea** (grade >=3): 2% (n=46); **increased ALT** (grade any): 47% (n=46); **increased ALT** (grade >=3): 4% (n=46); **fatigue** (grade any): 45% (n=46); **fatigue** (grade >=3): 4% (n=46); **thrombocytopenia** (grade any): 39% (n=46); **thrombocytopenia** (grade >=3): 4% (n=46); **dysgeusia** (grade any): 37% (n=46)
- **Efficacy:PFS_rate** 59.5%; **PFS** 13.1 months; **OS** 23.5 months; **ORR** 41.7%; **DoR** 17.3 months

rucaparib 600 mg PO BID maintenance — gated on germline or somatic BRCA1/2 or PALB2 + platinum induction (Reiss 2021 regimen)

Reiss et al. *Journal of Clinical Oncology* 2021

- **Disease context:Platinum-sensitive advanced PDAC with germline or somatic BRCA1/2 or PALB2 pathogenic variant** (maintenance) — 46 enrolled, 42 evaluable; germline BRCA1 n=7, BRCA2 n=27, PALB2 n=6; somatic BRCA2 n=2. Broader than POLO by including PALB2 and somatic variants.
- **n:** 46
- **Toxicities:anemia** (grade any): 74% (n=46); **anemia** (grade >=3): 22% (n=46); **nausea** (grade any): 48% (n=46); **nausea** (grade >=3): 2% (n=46); **increased ALT** (grade any): 47% (n=46); **increased ALT** (grade >=3): 4% (n=46); **fatigue** (grade any): 45% (n=46); **fatigue** (grade >=3): 4% (n=46); **thrombocytopenia** (grade any): 39% (n=46); **thrombocytopenia** (grade >=3): 4% (n=46); **dysgeusia** (grade any): 37% (n=46)
- **Efficacy:PFS_rate** 59.5%; **PFS** 13.1 months; **OS** 23.5 months; **ORR** 41.7%; **DoR** 17.3 months